DADT-D-0012-26-EVENING · DATA SCIENCE & VISUALIZATION

Thai Highway Accidents 2021–2025

When and where should checkpoint labor be positioned to prevent alcohol-related road deaths?

Traffic Enforcement Resource Allocation Manager · Royal Thai Police / DDPM

110,434
Crashes

13,901
Deaths

5 years
Jan 2021 – Dec 2025

77
Provinces

Thailand loses about 50 people a day on its roads

25.4
Road deaths per 100,000 people (2021)

18,218
Estimated road deaths a year, all roads

9th
Highest rate of 175 WHO member countries

83.8%
Of them motorcyclists — about one in five globally

Why our own totals are smaller — and not in conflict

WHO counts every road in Thailand and adjusts for under-reporting. **This project analyses the 110,434 highway crashes and 13,901 deaths recorded by the Ministry of Transport between 2021 and 2025** — a recorded subset of the national toll, not a contradiction of it.

Source: World Health Organization, Global status report on road safety 2023 — Thailand country profile, data year 2021. This is the only figure in the deck that does not come from Datasource/.

What this dataset can answer, and what it cannot

110,434
Crashes, after removing 1,702 duplicate records

99.4%
Carry usable coordinates

100%
Dates and times parsed, all five years

160
Road sections with matched traffic volume

The limitation, stated before anyone has to ask

This file covers about 17% of Thailand's road deaths

2,780 deaths a year here against 17,447 nationally (Department of Disease Control three-database system, 2024)

It records highways, rural highways and expressways only — not city streets or local roads, which is where most motorcyclists die

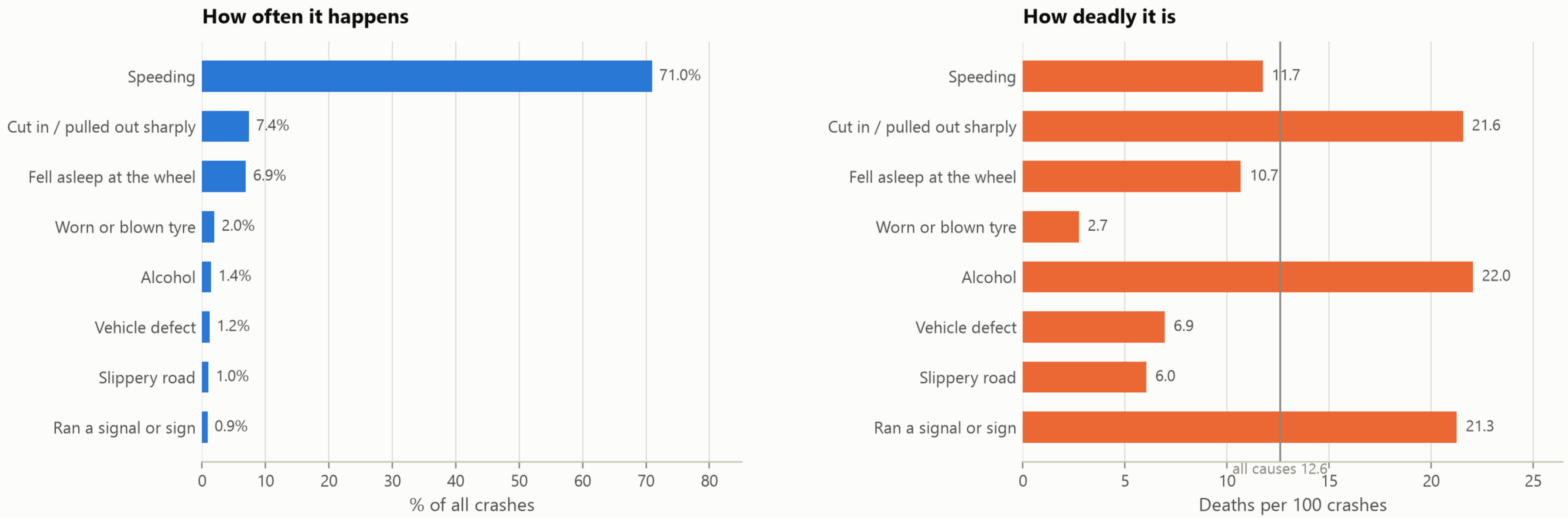
Method: risk is measured against exposure — crashes per vehicle-kilometre, using traffic volume × section length × years — so a busy road is not automatically a dangerous one. Every conclusion that follows is about highways, not about Thailand as a whole: motorcycles are 36% of deaths in this file against roughly 81% nationally.

PART 1 · DESCRIPTIVE

Alcohol is in the deadliest tier of crash causes

Cause: the most common is not the most lethal

Speeding is 71% of crashes but kills below the average rate — alcohol, cutting in and running signals are close to twice as deadly



Source: Highway accident records, Office of the Permanent Secretary, Ministry of Transport, 2021–2025 (110,434 crashes after removing 1,702 duplicates)

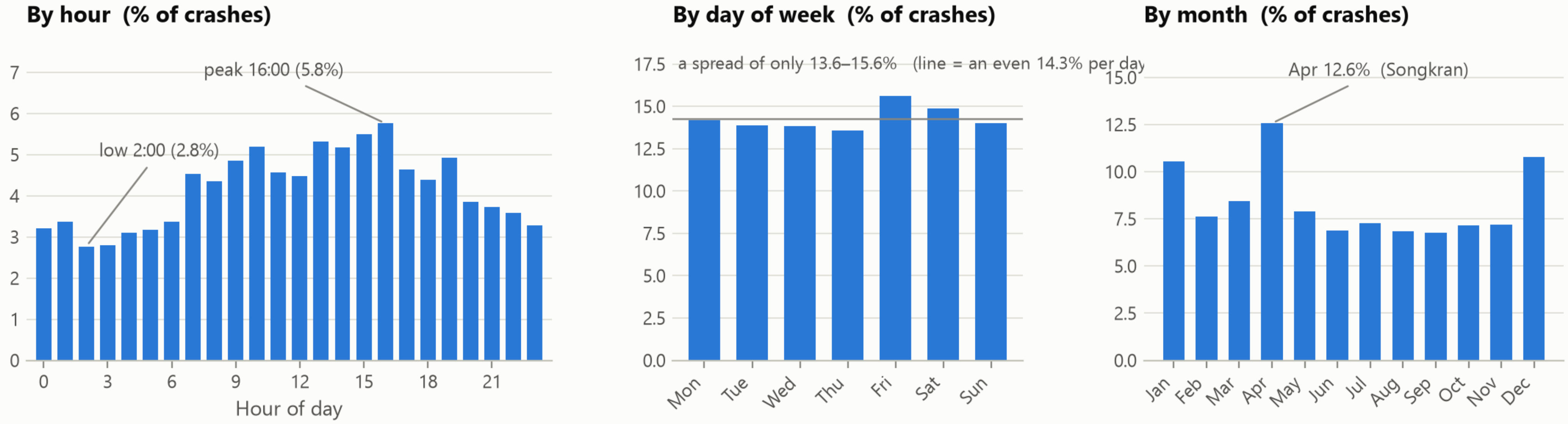
Alcohol is 1.4% of recorded crashes but kills 22.0 per 100 — against 12.6 across all crashes and 11.7 for speeding, which is 71% of them. Three other causes sit statistically level with it, so the point is the magnitude, not the ranking.

PART 1 · DESCRIPTIVE

Time of day decides deployment; day of week does not

When crashes happen

Hours differ by nearly 3x, but days of the week barely differ at all — checkpoints should be timed by hour, not by day



Source: Highway accident records, Office of the Permanent Secretary, Ministry of Transport, 2021–2025 (110,434 crashes after removing 1,702 duplicates)

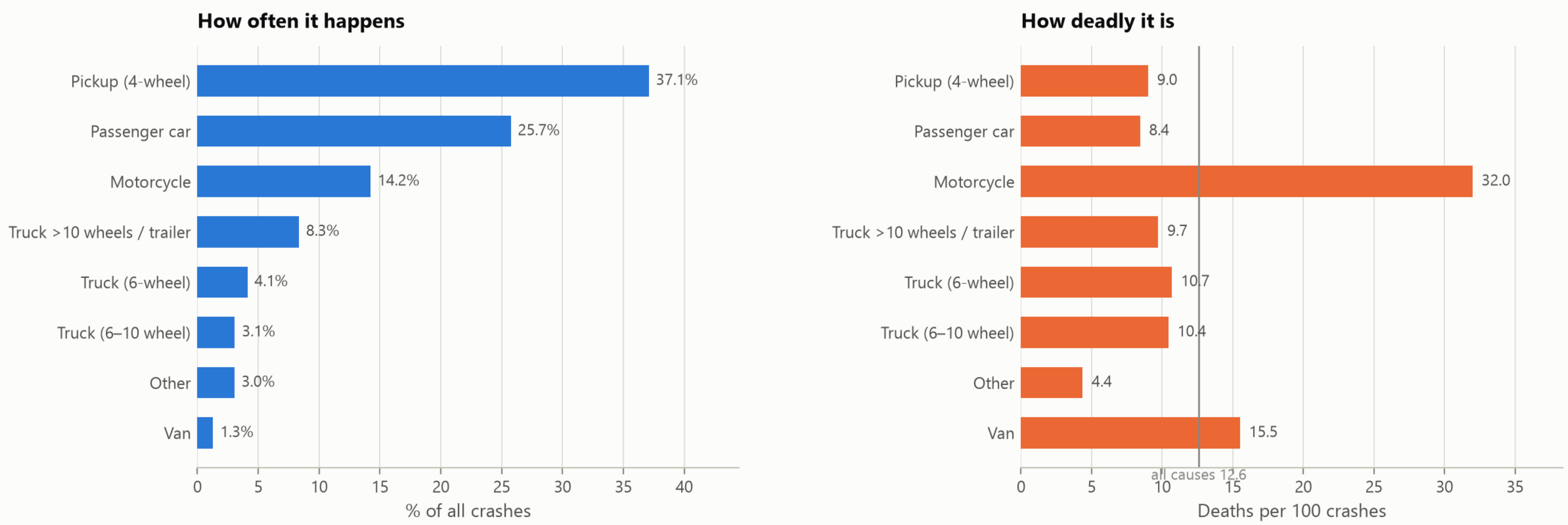
Hours differ by 2.1× — 02:00 carries 2.8% of crashes and 16:00 carries 5.8% — while weekdays sit inside a 13.6–15.6% band, near an even 14.3% split. Alcohol is the exception this chart does not isolate: 53% of alcohol-related crashes fall on Friday–Sunday and 59% between 19:00 and 04:00.

PART 1 · DESCRIPTIVE

Motorcycles crash less often and kill far more per crash

First vehicle involved

Pickups and passenger cars crash most often, but motorcycles kill several times more people per crash



Source: Highway accident records, Office of the Permanent Secretary, Ministry of Transport, 2021–2025 (110,434 crashes after removing 1,702 duplicates)

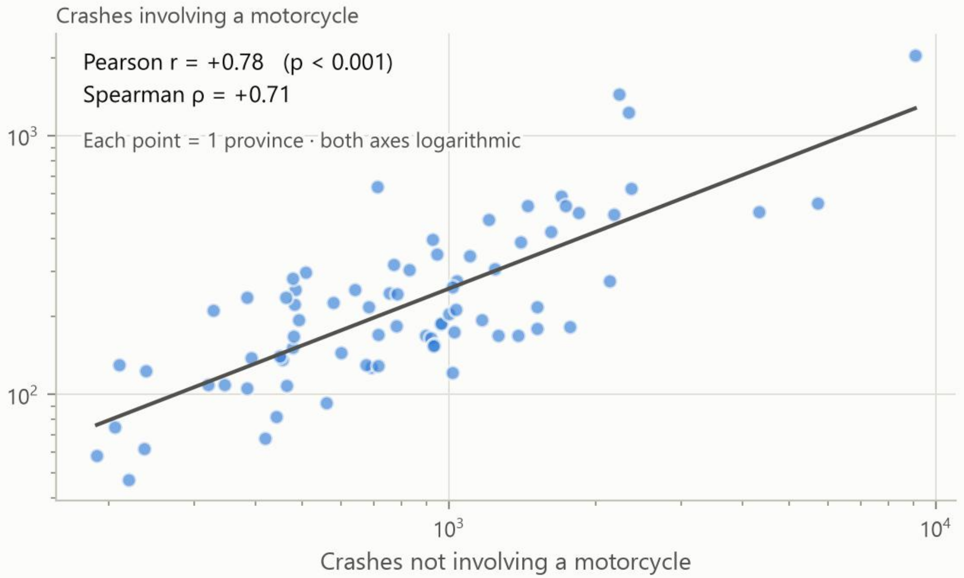
Pickups are 37% of crashes and kill 9.0 per 100. Motorcycles are 14% of crashes and kill 32.0 — 3.6× more per crash. Road alignment and weather point the same way: straight road is 71.5% of crashes at 12.3 deaths per 100, no safer than a curve; and 86% of crashes happen in clear weather, which is deadlier per crash than rain — 13.1 against 7.9.

The strongest link, and a link that is not there

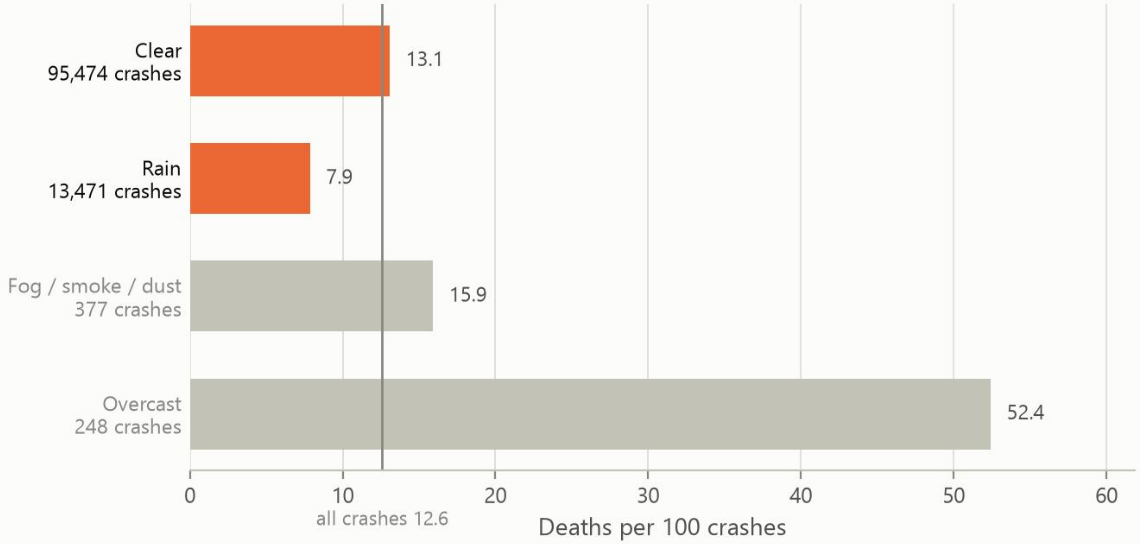
Crash risk is predictable — but weather is not why crashes turn deadly

Left: motorcycle and non-motorcycle crashes rise together across all 77 provinces (r = +0.78). Right: weather explains 0.2% of the variation in deaths per crash

A strong relationship



No meaningful relationship

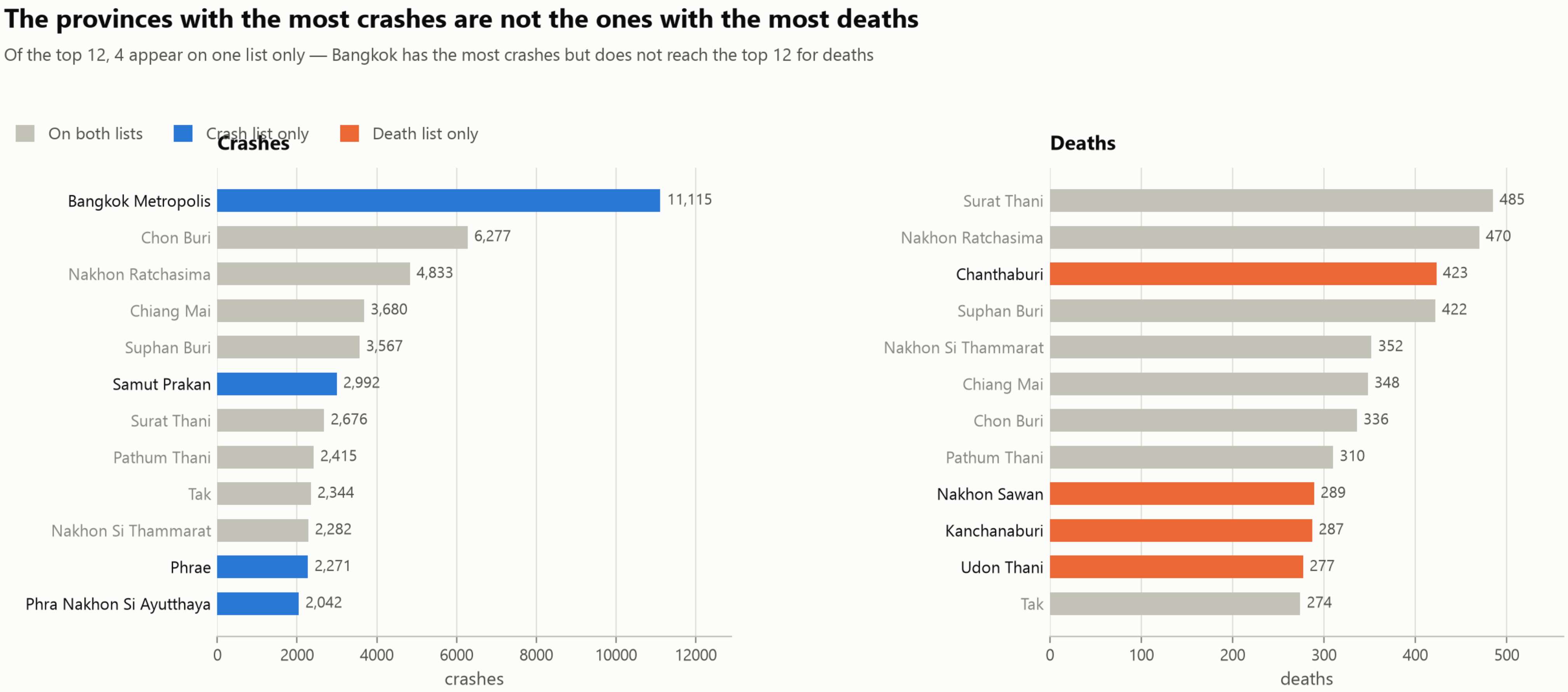


Clear and rain are 98.7% of all crashes, and weather explains 0.2% of the variation in deaths per crash (η = 0.04) — rain is the less deadly of the two.
Fog and overcast are greyed: together under 0.6% of crashes, and overcast is recorded on only 248 of them, falling every year — a reporting pattern, not a hazard.

Source: Highway accident records, Office of the Permanent Secretary, Ministry of Transport, 2021–2025 (110,434 crashes after removing 1,702 duplicates)

Left: motorcycle and non-motorcycle crashes rise together across all 77 provinces — r = +0.78, Spearman +0.71, p < 0.001. Right: weather explains 0.2% of the variation in deaths per crash (η = 0.04), and rain is the less deadly of the two large categories — 7.9 deaths per 100 against 13.1 for clear.

Crash hot spots are not death hot spots



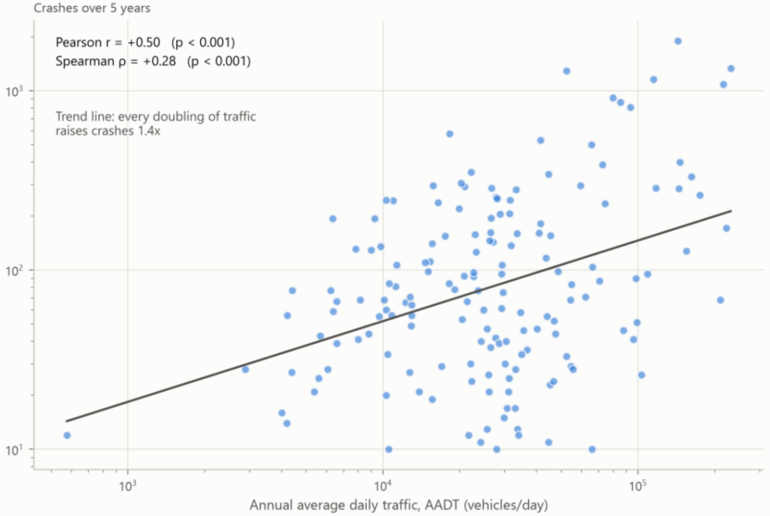
Source: Highway accident records, Office of the Permanent Secretary, Ministry of Transport, 2021–2025 (110,434 crashes after removing 1,702 duplicates)

Of the top 12 on each list, only 8 appear on both. Bangkok has the most crashes — 11,115 — but does not reach the top 12 for deaths: 2.1 deaths per 100 crashes against 12.6 nationally. Allocating enforcement by crash count sends people to the wrong provinces.

Traffic volume predicts how many crashes, not how deadly

More traffic means more crashes — but far more weakly than expected

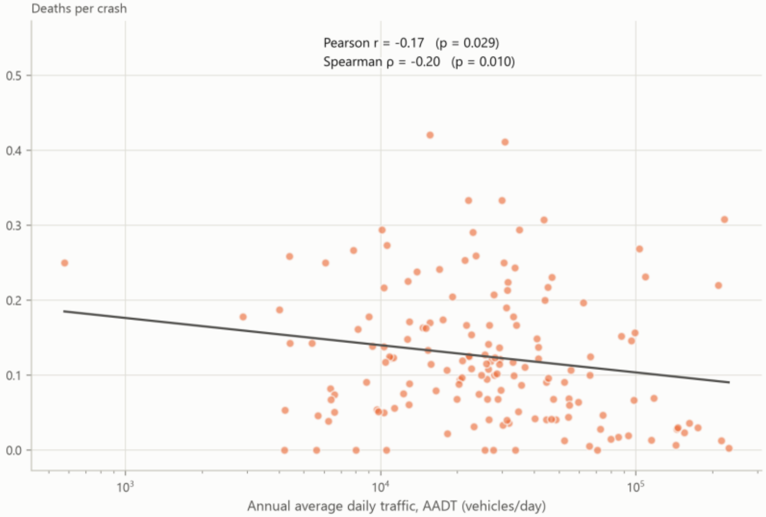
The direction is clearly positive but the spread is wide: at the same AADT, crash counts vary by up to 50x, so something other than volume is at work
Each point = one survey section · both axes logarithmic



Source: crash records, Office of the Permanent Secretary, Ministry of Transport, with AADT from the Bureau of Highway Safety, Department of Highways, 2021–2025 · 160 survey sections on 6 main highways (26,027 crashes)

But busier roads kill fewer people per crash

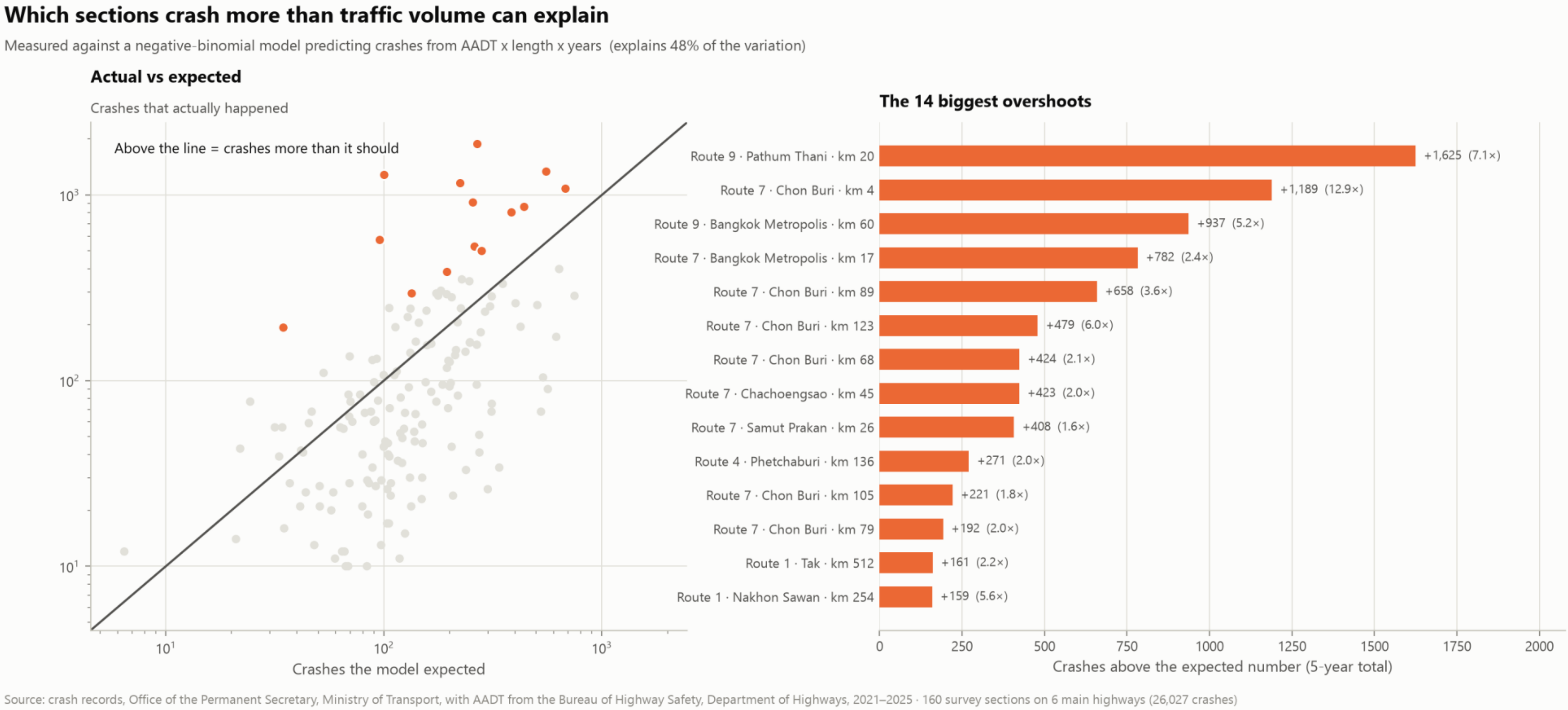
The relationship reverses and is very weak — traffic volume predicts how many crashes happen, not how deadly they are



Source: crash records, Office of the Permanent Secretary, Ministry of Transport, with AADT from the Bureau of Highway Safety, Department of Highways, 2021–2025 · 160 survey sections on 6 main highways (26,027 crashes)

Left: AADT against crash count — Pearson +0.50, Spearman +0.28. Positive, but weaker than expected: at the same traffic volume, crash counts differ by up to 50×. Right: AADT against deaths per crash — −0.17 and −0.20, reversed and very weak. This pair covers the six main Department of Highways routes only: 160 survey sections and 26,027 crashes, 23.6% of the file, because AADT exists only for those routes.

Crashing more than traffic volume explains



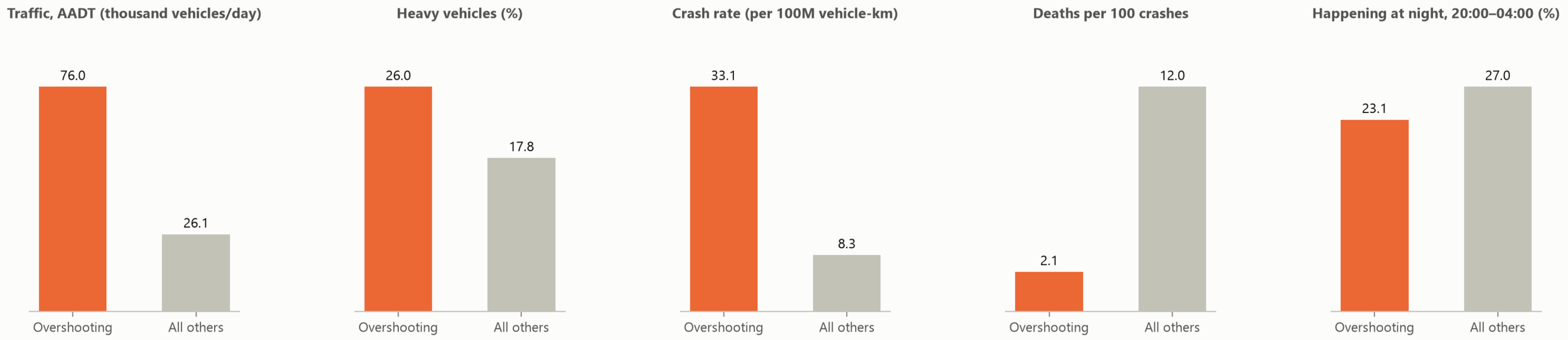
Five archetypes, one ranking. k-means on traffic volume, heavy-vehicle share, crash rate and deaths per crash splits the 160 sections into five kinds: freight corridors carry 29% of the deaths in this sample and killer highways 25%, while urban motorways carry 14% and kill least per crash. Ranked against a negative-binomial model that explains 48% of the variation in crash counts, the worst section runs 12.9× the crashes it should.

PART 1 · EXCESS RISK

What the worst sections have in common

What the over-crashing sections have in common

The 14 biggest overshoots — 79% are motorway (Routes 7 and 9): dense traffic, frequent crashes, but several times fewer deaths per crash than elsewhere



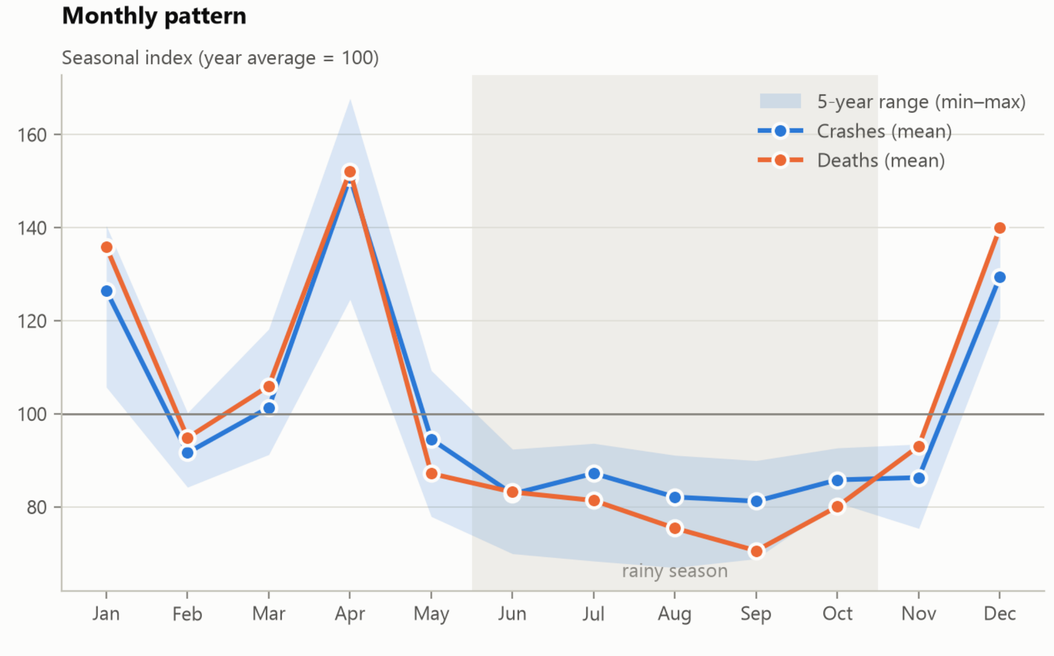
Source: crash records, Office of the Permanent Secretary, Ministry of Transport, with AADT from the Bureau of Highway Safety, Department of Highways, 2021–2025 · 160 survey sections on 6 main highways (26,027 crashes)

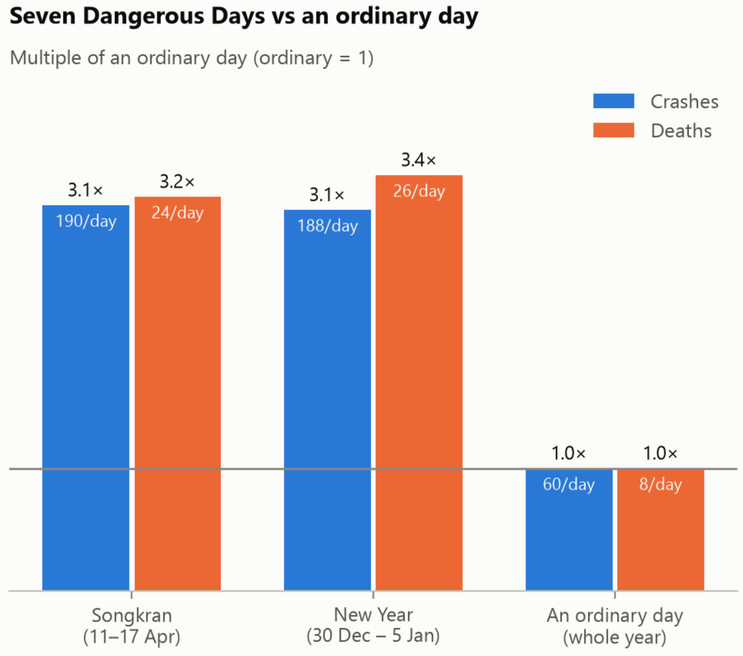
79% of the 14 biggest overshoots are motorway — Routes 7 and 9. They carry 76,000 vehicles a day against 26,100 elsewhere and crash four times as often (33.1 against 8.3 per 100 million vehicle-kilometres), yet kill 2.1 per 100 crashes against 12.0. The worst sections share a road type; they are not a scatter of unlucky places.

The spike is real — and partly measures enforcement

Seasonality — festivals matter far more than the rainy season

Index 100 = the yearly average · April runs 51% above it, while the rainy season runs below it



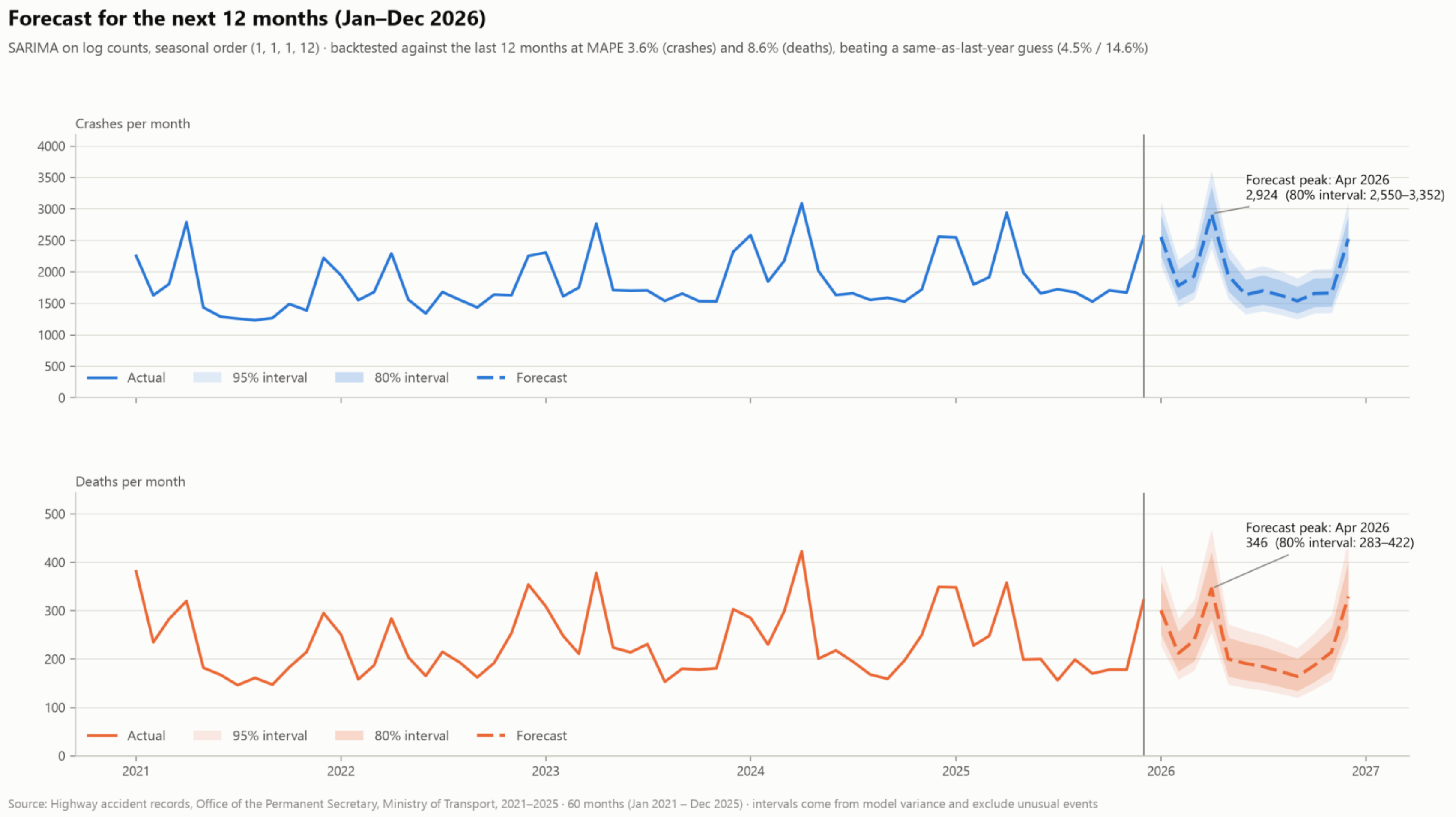


Source: Highway accident records, Office of the Permanent Secretary, Ministry of Transport, 2021–2025 · 60 months (Jan 2021 – Dec 2025)

Crashes run 3.1× an ordinary day inside the Seven Dangerous Days — Songkran 11–17 April at 3.14× and New Year 30 December – 5 January at 3.11×, level with each other.

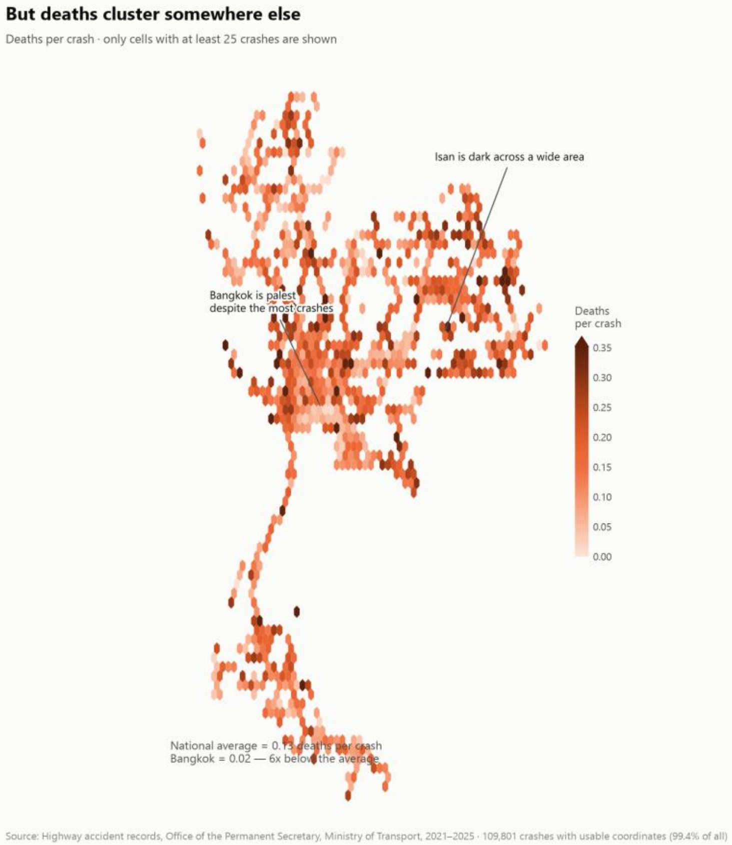
Read the alcohol share with care: 5.8% of crashes inside those windows are recorded as alcohol-related against 0.8% outside — about seven times — because that is when officers are running breath tests. The campaign window is when Thailand measures drink-driving, not only when it happens. April runs 51% above the yearly average and September 19% below it: festivals drive the year, not the rainy season, which runs below average.

Planning next year's checkpoint capacity



SARIMA on log counts, backtested against the last 12 months at MAPE 3.6% for crashes and 8.6% for deaths — better than assuming the same as last year. April 2026 is forecast at 2,924 crashes (80% interval 2,550–3,352) and 346 deaths (283–422); September is the trough at 1,539. Reporting lag was tested and rejected: December 2021 had 551 days to be reported and 2,222 crashes, December 2025 had 28 days and 2,560.

Deploy where deaths are, not where crashes are



The deployment plan

- Target sections, not provinces. The 14 sections crashing furthest above their expected rate are 79% motorway, Routes 7 and 9 — while deaths per crash concentrate in the north-east, where Bangkok's crash volume never reaches.
- Staff by month, not by annual average. April needs 1.9× September's capacity: 2,924 forecast crashes against 1,539.
- Surge for the Seven Dangerous Days at roughly 3× ordinary staffing — Songkran 11–17 April and New Year 30 December – 5 January, sized equally, because the two windows are level.
- Start campaigns two weeks early. The rise begins before the festival week itself: mid-March for Songkran, mid-December for New Year.
- Set checkpoints by hour, not by day. Weekdays differ by under two points; hours differ by 2.1×. For alcohol specifically, Friday–Sunday and 19:00–04:00 carry most of it.

How to run it, and what it cannot see

Implementation

- Tailor by archetype: freight corridors need heavy-vehicle enforcement, killer highways need speed and alignment work, and urban motorways need faster incident response rather than more checkpoints.
- Measure against the forecast's lower bound. If April 2026 comes in below 2,550 crashes, the intervention beat the trend rather than the weather.
- Report deaths per 100 crashes by section, not crash counts — crash counts fall whenever traffic falls, which flatters any campaign.

Limitation — and what follows from it

Alcohol is recorded on 1.4% of crashes, but almost entirely inside the 14 campaign days each year. Outside them it appears only when a crash was serious enough to investigate — the fatality premium falls from 2.31× to 1.17× once detection is consistent, and inverts at night. Detected year-round at the in-window rate, there would be roughly 4,900 more alcohol crashes and 750 more alcohol deaths than the 345 this data records. **Year-round breath testing is therefore a prerequisite for evaluating any deployment plan, including this one. Without it, a fall in recorded alcohol crashes cannot be told apart from a fall in testing.**

Two further limits: this file is highways only — about 17% of Thailand's road deaths — and the exposure analysis covers six main routes, 23.6% of the crashes. Treat the plan as a hypothesis with a measurable test attached, not as a settled answer.

REFERENCES

Sources, geography and tools

Data

- Highway accident records 2021–2025 (B.E. 2564–2568) — Office of the Permanent Secretary, Ministry of Transport. Every number in this deck except the WHO figures comes from this file.
- Annual Average Daily Traffic 2021–2025 — Department of Highways.
- Annual Average Daily Traffic 2021–2025 — Department of Rural Roads.
- Thailand's national road-death total — Department of Disease Control, three-database integrated system, 2024.
- World Health Organization, Global status report on road safety 2023 — Thailand country profile, data year 2021. who.int/thailand/our-work/road-safety

Geography and tools

- Highway centrelines — ArcGIS open data, Department of Highways and Department of Rural Roads. Used as map geometry only, never as an analytical value.
- Province boundaries — GADM 4.1. Map geometry only.
- Python 3.14 with pandas, statsmodels, scikit-learn, scipy and matplotlib; MapLibre GL JS for the maps; PptxGenJS for this deck.
- Presentation site (three scrolling tabs, one per presenter): https://<github-user>.github.io/thailand-road-accidents/